Supporting Information

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SI Text

References from which the phylogeny for phylogenetically independent contrasts was assembled:

Marsupial orders and Diprotodontian families: Beck 2008 (1)
Positioning of *Acrobates*: Osborne and Christidis 2002 (2); Kavanagh 2004 (3)
Potoroids: Westerman et al. 2004 (4)
Genus *Macropus*: Westerman et al. 2002 (5)
Burramyidae: Osborne and Christidis 2002 (6)
Petauroidea: Osborne and Christidis 2001 (7)
Peramelemorpha: Westerman et al. 1999 (8)
Dasyuromorpha: Krajewski et al. 2000 (9)
Most placental relationships: Bininda-Emonds et al. 2007 (10)
Afrotheria: Asher et al. 2003 (11)
Carnivora: Flynn et al. 2005 (12)
Muroid rodents: Steppan et al. 2004 (13)

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2. Osborne MJ, Christidis L, Norman JA (2002) Molecular phylogenetics of the Diprotodontia (kangaroos, wombats, koala, possums, and allies). *Mol Phylogenet Evol* 25:219–228.
3. Kavanagh JR, Burk-Herrick A, Westerman M, Springer MS (2004) Relationships among families of Diprotodontia (Marsupialia) and the phylogenetic position of the autapomorphic honey possum (*Tarsipes rostratus*). *J Mamm Evol* 11:207–222.
4. Westerman M, Loke S, Springer MS (2004) Molecular phylogenetic relationships of two extinct potoroid marsupials, *Potorous platyops* and *Caloprymnus campestris* (Potoroinae: Marsupialia). *Mol Phylogenet Evol* 31:476–485.
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7. Osborne MJ, Christidis L (2001) Molecular phylogenetics of Australo-Papuan possums and gliders (family Petauridae). *Mol Phylogenet Evol* 20:211–224.
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9. Krajewski C, Wroe S, Westerman M (2000) Molecular evidence for the pattern and timing of cladogenesis in dasyurid marsupials. *Zool J Linn Soc* 130:375–404.
10. Bininda-Emonds ORP, et al. (2007) The delayed rise of present-day mammals. *Nature* 446:507–512.
11. Asher RJ, Novacek MJ, Geisler JH (2003) Relationshiops of endemic African mammals and their fossil relatives based on morphological and molecular evidence. *J Mamm Evol* 10:131–194.
12. Flynn JJ, Finarelli JA, Zehr S, Hsu J, Nedbal MA (2005) Molecular phylogeny of the Carnivora (Mammalia): Assessing the impact of increased sampling on resolving enigmatic relationships. *Syst Biol* 54:317–337.
13. Steppan SJ, Adkins R, Anderson J (2004) Phylogeny and divergence date estimates of murid rodents based on multiple nuclear genes. *Syst Biol* 53:533–553.

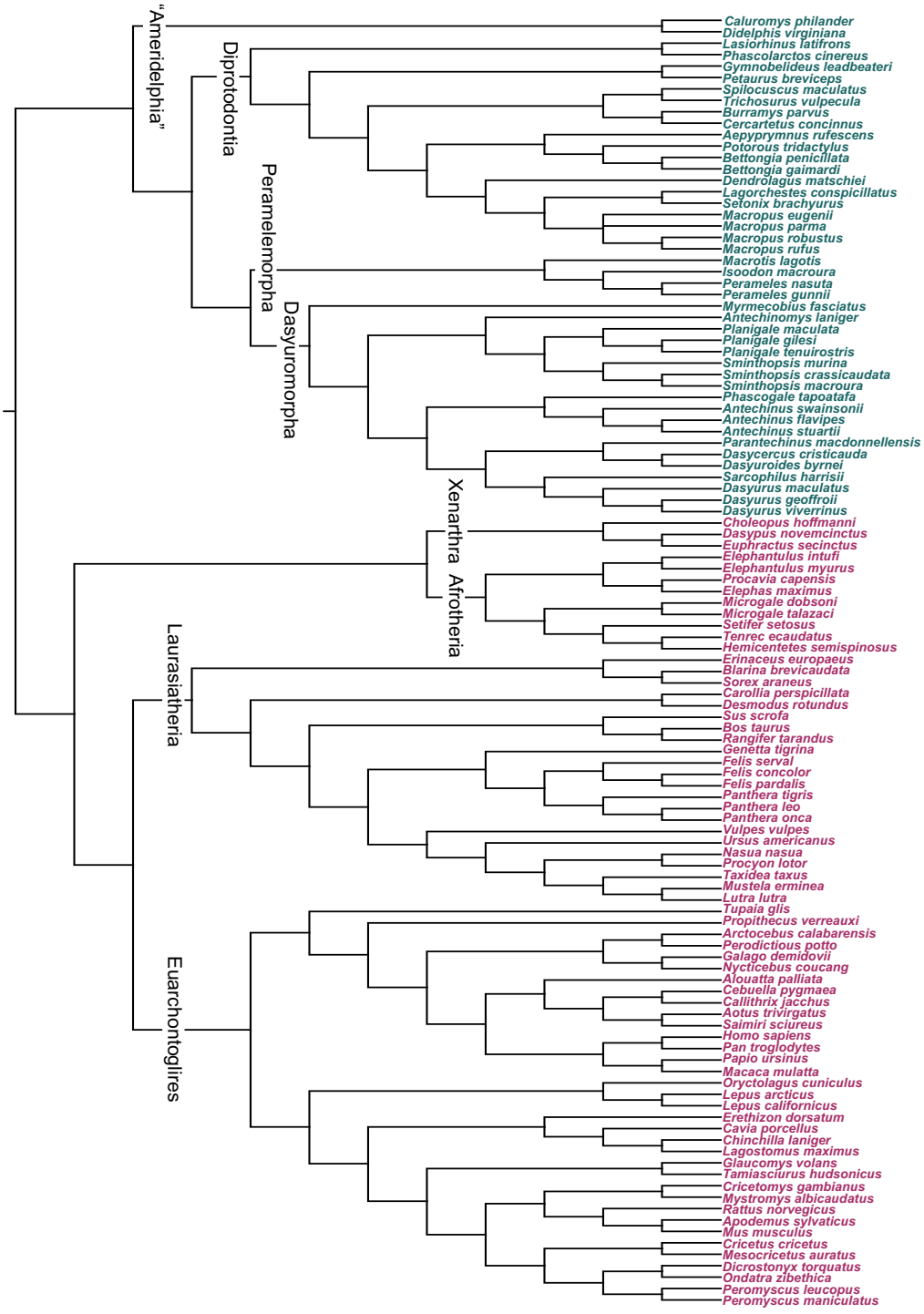


Fig. S1. Phylogeny used for phylogenetically independent contrasts analysis of partial correlation and bivariate regression analyses. The main Therian clades (ordinal for marsupials, superordinal for placentals) are identified.

Table S1. *P* values, Wilcoxon's rank-sum test (Holm-adjusted for multiple comparisons) of EQs among mammalian superorders

	Dasyuromorpha	Diprotodontia	Peramelemorpha	"Ameridelphia"	Laurasiatheria	Xenarthra	Afrotheria
Dasyuromorpha							
Diprotodontia	**0.000**						
Peramelemorpha	**0.000**	**0.005**					
"Ameridelphia"	1	1	**0.023**				
Laurasiatheria	1	0.000	0.000	1			
Xenarthra	1	1	1	1	1		
Afrotheria	1	1	0.011	1	1	1	
Euarchontoglires	0.001 (1)	0.000 (0.056)	0.000 (0.000)	0.191 (1)	**0.000** (1)	0.23 (1)	**0.015** (1)

P values are shown. Significant differences are in bold type. Shaded areas represent comparisons between marsupial and placental superorders. Numbers in parentheses for tests involving Euarchontoglires are comparisons in which Primates was removed from Euarchontoglires. The correlation between Euarchontoglires excluding Primates and Diprotodontia is just above the significance threshold.

Table S2. Partial correlations of residuals of brain size, gestation, weaning age, litter size and BMR in marsupials and placentals

	Brain size	Gestation	Weaning age	Litter size	BMR
Marsupials					
Brain size	1	−0.10	**0.38***	**−0.45****	−0.03
Gestation	0.01	1	0.20	−0.22	−0.06
Weaning age	**0.40****	**0.36***	1	−0.10	−0.21
Litter size	**−0.59*****	−0.00	−0.02	1	−0.03
BMR	0.03	0.04	−0.25	−0.14	1
Placentals					
Brain size	1	−0.11	**0.27***	**−0.41****	**0.32****
Gestation	−0.07	1	**0.67*****	**−0.45*****	−0.14
Weaning age	**0.48*****	**0.48*****	1	0.22	0.04
Litter size	**−0.30***	**−0.55*****	−0.00	1	0.01
BMR	**0.27***	**−0.46*****	0.13	−0.07	1

Gestation, weaning age, and litter size residuals are derived from log-log regression against body mass published with brain weights. Values above diagonals represent correlations of phylogenetically independent contrasts; values below diagonals are noncorrected. Asterisks indicate the level of significance: $0.1 > P > 0.05$; *$0.05 > P > 0.01$; **$0.01 > P > 0.001$; ***$P < 0.001$.

Table S3. Partial correlations of body size–adjusted residuals of brain size gestation, weaning age, litter size, and BMR in marsupials and placentals

	Brain size	Gestation	Weaning age	Litter size	BMR
Marsupials					
Brain size	1				
Gestation	0.03	1			
Weaning age	**0.36***	**0.36***	1		
Litter size	**−0.56*****	0.00	−0.03	1	
BMR	−0.00	0.03	−0.23	−0.17	1
Placentals					
Brain size.	1				
Gestation	−0.08	1			
Weaning age	**0.45*****	**0.49*****	1		
Litter size	**−0.35****	**−0.55*****	0.01	1	
BMR	**0.26***	**−0.46*****	0.14	−0.08	1

The analysis is as in Table S2, but gestation, weaning age, and litter size residuals are derived from regression against body mass published with BMR weight. Asterisks indicate the level of significance: $0.1 > P > 0.05$; *$0.05 > P > 0.01$; **$0.01 > P > 0.001$; ***$P < 0.001$.

Table S4. Partial correlations of body size–adjusted residuals of brain size, weaning age, litter size, and BMR in a larger dataset of marsupials

	Brain size	Weaning age	Litter size	BMR
Brain size	1	**0.36***	**−0.55*****	−0.07
Weaning age	**0.40*****	1	−0.09	−0.11
Litter size	**−0.57*****	−0.08	1	−0.15
BMR	0.05	−0.13	−0.13	1

Leaving out gestation allowed for the inclusion of seven additional species. Values above diagonals represent correlations of phylogenetically independent contrasts; values below diagonals are noncorrected, log10-transformed data. Asterisks indicate the level of significance: 0.1 > P > 0.05; *0.05 > P > 0.01; **0.01 > P > 0.001; ***P < 0.001.

Table S5. Bivariate regressions of brain size residuals against residuals of the maternal investment parameters included in the partial regression to assess the variation in brain size explained by each variable

	Slope	SE	R^2	P
Marsupials				
Gestation	0.105	0.098	0.026	0.288
Weaning age	0.268	0.102	0.138	0.012*
Litter size	−0.284	0.058	0.389	0.000***
Placentals				
Gestation	0.415	0.028	0.212	0.000***
Weaning age	0.559	0.072	0.481	0.000***
Litter size	−0.496	0.082	0.361	0.000***

The analysis is as in Table 1, but gestation, weaning age, and litter size residuals are derived from regression against body mass published with BMR values.

Table S6. Parameters used in the partial correlation and residual regression analyses of marsupials

	Body weight (brain), g	Brain weight, g	Gestation, d	Lactation, d	Litter size	Body size (BMR), g	BMR, KJ/h
Dasycercus cristicauda	70.2	1.24	34.5	107	5.9	88.8	0.93
Dasyuroides byrnei	98.4	1.5	34	118	4.3	118.2	1.66
Dasyurus geoffroii	760	5.17	16	160	6	1,354	11.42
Dasyurus maculates	2,599	9.41	21	134	5.3	1,782	11.81
Dasyurus viverrinus	813	5.33	20.5	159	6	910	8.22
Parantechinus macdonnellensis	24	0.49	50.5	98	5.9	43.1	0.55
Sarcophilus harrisii	7,430	16.09	31	180	2.9	5,050	28.39
Antechinus flavipes	30.7	0.64	26.5	90	8.4	46.5	0.91
Antechinus stuartii	29.8	0.76	27	90	7.2	36.5	0.73
Antechinus swainsonii	63.5	1.09	28	90	8	66.9	1.28
Phascogale tapoatafa	150	1.81	28.5	151	6.7	157	2.55
Planigale gilesi	6.1	0.17	15.8	75	6.1	9.4	0.26
Planigale maculata	6.7	0.2	19.5	70	7.7	13.1	0.27
Planigale tenuirostris	3.9	0.14	19	95	6	7.1	0.23
Antechinomys laniger	17.8	0.47	12	96	6	24.2	0.48
Sminthopsis crassicaudata	13.3	0.36	14.5	68	7.4	17.7	0.44
Sminthopsis macroura	17	0.4	11.8	70	6.5	22	0.47
Sminthopsis murina	15.9	0.44	12.5	65	8	19	0.43
Myrmecobius fasciatus	405.4	4.25	14	270	4	480	3.73
Lasiorhinus latifrons	19,264	55.5	21	333	1	25,000	50.2
Phascolarctos cinereus	5,888	20.2	34.5	365	1	4,765	21.05
Burramys parvus	36.6	1	14.8	67	3.6	44.3	0.74
Cercartetus concinnus	9.5	0.3	59	50	4.6	18.6	0.45
Gymnobelideus leadbeateri	90.5	2.85	17	115	1.5	166	2.07
Petaurus breviceps	89.18	2.09	16.5	120	2.3	128.1	1.78
Tarsipes rostratus	9	0.36	24	82	2.6	10	0.58
Spilocuscus maculatus	3,127	13.4	13	240	1	4,250	22.19
Trichosurus vulpecula	1,902	11	17.3	185	1	1,982	12.74
Aepyprymnus rufescens	1,693	12.3	22.5	163	1	2,820	21.52
Bettongia gaimardi	1,273	11.7	21	155	1	1,385	12.88
Bettongia penicillata	981	9.56	21	165	1	1,070	11.07
Dendrolagus matschiei	4,160	27.5	44	500	1	6,960	29.35
Lagorchestes conspicillatus	2,015	14	30	204	1	2,260	17.09
Macropus eugenii	4,425	23.7	29	325	1	4,796	27.93
Macropus parma	3,557	18.7	33	294	1	3,750	17.09
Macropus robustus	35,540	59	34	409	1	29,300	111.79
Macropus rufus	44,841	62.4	33	378	1	32,490	116.13
Potorous tridactylus	1,283	9.53	38	163	1	1,120	10.12
Setonix brachyurus	1,931	13.9	27	310	1	2,510	16.13
Isoodon macrourus	822	4.42	12.5	60	3.3	1,551	12.77
Perameles gunnii	1,002	4.5	12.3	58	2.4	837.3	8.44
Perameles nasuta	1,273	4.96	12.5	66	2.7	645	6.35
Macrotis lagotis	1,859	6.57	14	111	1.7	1,011	7.11
Caluromys derbianus	289	3.4	24	120	4.3	357	4.09
Didelphis virginiana	2,797	6.72	13	98	7.7	3,257	21.58
Didelphis marsupialis	2,842	6.35	—	94.3	6	1,329	12.28
Acrobates pygmaeus	8.9	0.35	—	97.5	2.8	14	0.3
Dasyurus hallucatus	513	3.38	—	142.5	6.4	584	5.98
Cercartetus lepidus	7.8	0.36	—	90	3.3	12.6	0.38
Cercartetus nanus	12	0.43	—	65	3.8	70	1.21
Petauroides volans	900	4.69	—	250	1	1,140	11.45
Isoodon obesulus	691	3.89	—	62.5	2.7	717.2	4.46